



Probabilistic modeling of wind energy potential for power grid expansion planning

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ABSTRACT

The increasing integration of wind energy generation into energy systems has led to difficulties in power flow calculations due to uncertainty and variability, which significantly affect the stability and reliability of power grids. Consequently, it is critical to evaluate the energy security limit of power grids based on probabilistic approaches. In this paper, we propose the probabilistic modeling of wind energy potential for power grid expansion planning. The proposed probabilistic model estimates wind energy potential through Weibull distribution, Monte Carlo Simulation (MCS), and the enhanced spatial modeling based on universal kriging. It can be used to establish the expansion plan for transmission and distribution facilities to resolve the variability and uncertainty issues of wind generation resources, which is expected to increase the penetration levels of renewable energy. To validate the proposed model, the empirical data from the Jeju Island's wind farms are considered in South Korea.

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1. Introduction

The percentage of renewable energy and renewable energy policies of each country are increasing due to the obligation to reduce greenhouse gas emissions according to the Paris Climate Change Accord, and the agreement to deplete fossil fuel [1]. According to Ref. [2], CO₂ emission is expected to increase by 2.2% annually by 2020, and existing fossil fuels such as oil and natural gas cause environmental pollution problems such as ozone layer depletion and climate change [3]. On the other hand, renewable energy sources are unlimited clean energy which emit none or very little amounts of air pollutants and greenhouse gases, e.g. solar energy, wind energy, biomass energy, hydropower, etc [2,4].

In 2018, renewable energy generation exceeded 33% of the global power capacity. Fig. 1 shows the percentage of estimated renewable energy share in global electricity production [5]. From 2017 to 2018, global renewable energy generation increased by approximately 14.5% from 2166.5 to 2480.4 TWh, respectively, and has changed primarily from solar and wind. During the same time period, the United States and Germany increased solar power by approximately 24.4% from 78.1 to 97.1 TWh and 17.2% from

39.4 TWh ('17) to 46.2 TWh, respectively, while France increased wind power by 15.8% from 24.3 to 28.2 TWh, respectively [6].

According to the 8th Basic Plan for Long-term Electricity Supply and Demand, the Republic of Korea aims to achieve 20% (132 TWh) of renewable energy generation by 2030. Considering the stability of the power system and the potential of renewable energy, the target for renewable energy power generation was set at 20%, while the amount of renewable energy capacity newly supplied by 2030 should be 48.7 GW. Fig. 2 shows the Republic of Korea's Renewable Energy 2016–2030 Implementation Plan.

“Carbon Free Island Jeju by 2030” is a project to achieve Korea's decarbonization and renewable energy generation goals, and aims to install 2350 MW of Jeju's onshore and offshore wind farm by 2030. Due to the geographical characteristics of Jeju Island, installation of offshore wind farm is more suitable than onshore, and if the conditions for offshore wind are well established, offshore wind farm has the advantage of higher electricity production than onshore [7]. We need to estimate wind energy potential and analyze power system operational efficiency before installing new offshore wind farms to achieve the Carbon Free Island goal, and it is necessary to consider transmission expansion plans according to the increase in HVDC and offshore wind farms [8–10]. China has expanded the capacity of wind power, but there is a problem of curtailment due to transmission inadequacy and minimum generation limits [11]. In addition, high penetration above 20–30% of

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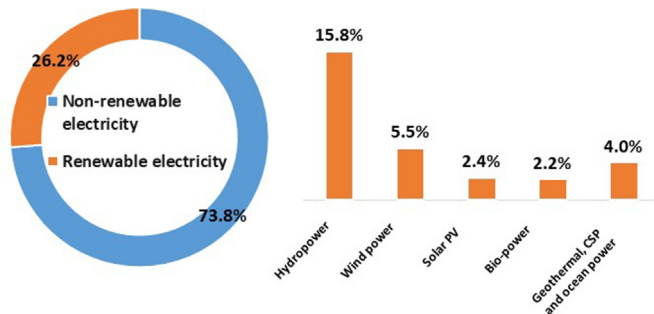


Fig. 1. Renewable energy share of global electricity production [1].

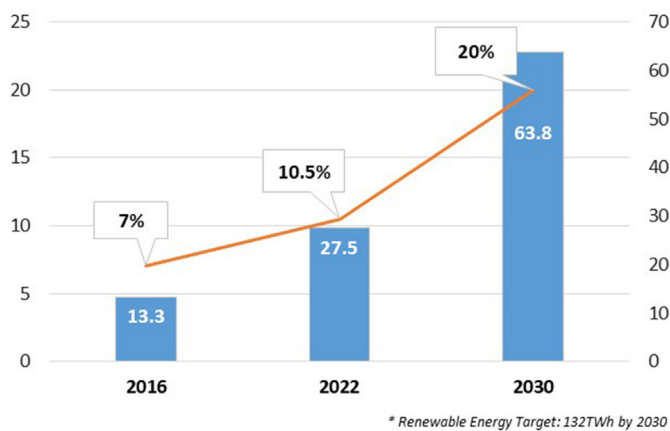


Fig. 2. South Korea's renewable energy target plan.

installed capacity cause system security problems [12–14]. Renewable energy sources centered on solar and wind power have problems of uncertainty and variability [15,16]. Specifically, wind power generation resources have greater variability because they are affected by weather factors such as temperature, wind speed and direction [17]. Wind power output is variable, although statistical patterns can be identified over a long period of time [18]. Therefore, in order to secure system reliability against high penetration of wind power generation, it is necessary to improve the existing power grid system analysis and probabilistic prediction techniques of renewable energy output or to develop new power generation methods [19–22]. In compliance with the criteria for expanding and stabilizing power systems, as well as efficient utilization of existing infrastructure or planning to expand the grid capacity as per the expansion of renewable energy, should be considered [23].

The Republic of Korea established a transmission system plan based on the Ministry of Trade, Industry, and Energy's Power System Reliability and Electrical Quality Maintenance Standards and is currently conducting continuous system reinforcement and expansion essential to maintain the power system reliability [24]. In many European countries, the rapid expansion of renewable energy has created a need to expand the grid [25]. In the future, long-term energy system modeling is required to secure high penetration of Renewable Energy and system reliability [26,27], and there is increasing awareness of the need to study grid expansion plans [28].

In this paper, we propose the probabilistic modeling of wind energy potential for power grid expansion planning. We analyzed the probabilistic model based on a Weibull distribution and Monte Carlo simulations using the measured data of existing wind farms

and the estimated potential capacity of wind power resources on Jeju Island, Korea, through the enhanced spatial modeling with universal kriging. The proposed algorithm will be a key a role to estimate the potential of wind generating resources for all spatio-temporal scales, and will be used to select the optimal location for new wind farm candidate sites through wind resource potential estimation. In addition, it can be applied to the establishment of a stable operation plan and expansion plan for transmission and distribution facilities with high wind power penetrations. The remainder of this paper is organized as follows. Section 2 presents the probabilistic modeling for estimating wind energy potential for new farm locations. In Section 3, the proposed probabilistic modeling of wind energy potential for power grid expansion planning is discussed. We present the probabilistic scenarios to use with the modeled wind power outputs, which reflects the Island's seasonal conditions, for the empirically modeled power grid for Jeju Island. Finally, conclusions are presented in Section 4.

2. Probabilistic modeling for estimating wind energy potential

In this section, the capacity factor [32] calculated from the wind farm output data fits a two-parameter Weibull distribution with shape (k) and scale (c), and we used the energy pattern method (EPM) [29–31] to calculate the two parameters. Sampled data were extracted from the results of the Monte Carlo simulations to calculate the scale and shape parameters. The 95% confidence intervals of the sample data were used to estimate the capacity of the new wind farm target points through universal kriging.

2.1. Probabilistic modeling based on Monte Carlo Simulation

Before the Monte Carlo simulation, the R program was used to select the probability distribution model suitable for the capacity factor data of the wind farm. Using a histogram of the Monte Carlo simulation results, the optimal probability model with the lowest Bayesian information criterion value was selected, and the applicability of the probability model was verified. Monte Carlo simulation is a method of random sampling based on parameters extracted by probability fitting of measured data. It is used to model the uncertainty of system elements and to evaluate the probabilistic steady-state stability of the power system [33].

We used a Weibull distribution function with two parameters, namely shape and scale, for wind power estimation [29]. In this section, the capacity factor was applied to the 2-parameter Weibull distribution function. The Weibull distribution function is expressed as

$$f(v) = \left(\frac{k}{c}\right) \left(\frac{v}{c}\right)^{k-1} \exp\left(-\left(\frac{v}{c}\right)^k\right) \quad (1)$$

where v is the wind power capacity factor, k is the shape parameter, and c is the scale parameter.

According to Refs. [34,35], dimensionless factor shape parameter(k) determines the shape of the curve. And scale parameter having same unit as wind power capacity factor. According to Ref. [36], for higher value of scale parameter the distribution will be spread over a wider range and having more probability of higher average wind power capacity factor. On the other hand, shape parameter shows the more probability of higher wind power capacity factor.

Parameter calculation of the Weibull distribution was applied using the EPM [29–31]. Eq. (2) is used to calculate the energy tendency factor, while Eqs. (3) and (4) represent the shape parameter k and scale parameter c of the Weibull distribution,

respectively, where $\bar{v}$ is the mean, and the definition of the gamma function used in scale parameter c is given in Eq. (5).

$$E_{pf} = \frac{\bar{v}^3}{(\bar{v})^3} \quad (2)$$

$$k = 1 + \frac{3.69}{(E_{pf})^2} \quad (3)$$

$$c = \frac{\bar{v}}{\Gamma\left(1 + \frac{1}{k}\right)} \quad (4)$$

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt \quad (5)$$

After calculating the Weibull distribution parameters, the sample data extraction and Monte Carlo simulations were performed 10,000 times to consider the correlation of the capacity factor between wind farms using universal kriging. Correlation between the wind farms should be considered because sampling without considering it can distort the probability characteristics of the total energy generation.

Eq. (6) is a function for calculating the two-parameter Weibull distribution mean, μ ; α and β are the shape and scale parameters of the sample data [37]. For the reliability function used in this study [38,39], the 95% confidence interval for the Monte Carlo simulation is calculated using Eq. (7).

$$\mu = \alpha \Gamma\left(1 + \frac{1}{\beta}\right) \quad (6)$$

$$R(v) = e^{-\left(\frac{v}{\alpha}\right)^\beta} \quad (7)$$

2.2. Estimation of capacity factors of wind farms based on spatial modeling

A variogram is characterized and shaped using the sill, nugget, and range parameters. They are applied to identify the spatial distribution characteristics of the data and are necessary to perform the kriging. A variogram can be defined as

$$2\gamma(d) = E\left[(z(x+d) - z(x))^2\right] \quad (8)$$

where x is the target point and d is the separation distance.

Theoretical variograms can have a variety of models, such as linear, spherical, exponential, and Gaussian. In this section, we used spherical models that are commonly used in spatial modeling [40]. The spherical model function can be written as

$$Sph = (s - n) \left[1.5 \left(\frac{d}{r} \right) - 0.5 \left(\frac{d}{r} \right)^3 \right] \quad (9)$$

where s is the sill, n is the nugget, and r is the range.

Kriging is a representative spatial modeling technique used in geostatistics. It uses weighted linear combinations of surrounding data that are already known to predict the characteristic data of the target point. In this paper, the capacity factor of the new wind farm was estimated using universal kriging. The weight was adjusted to

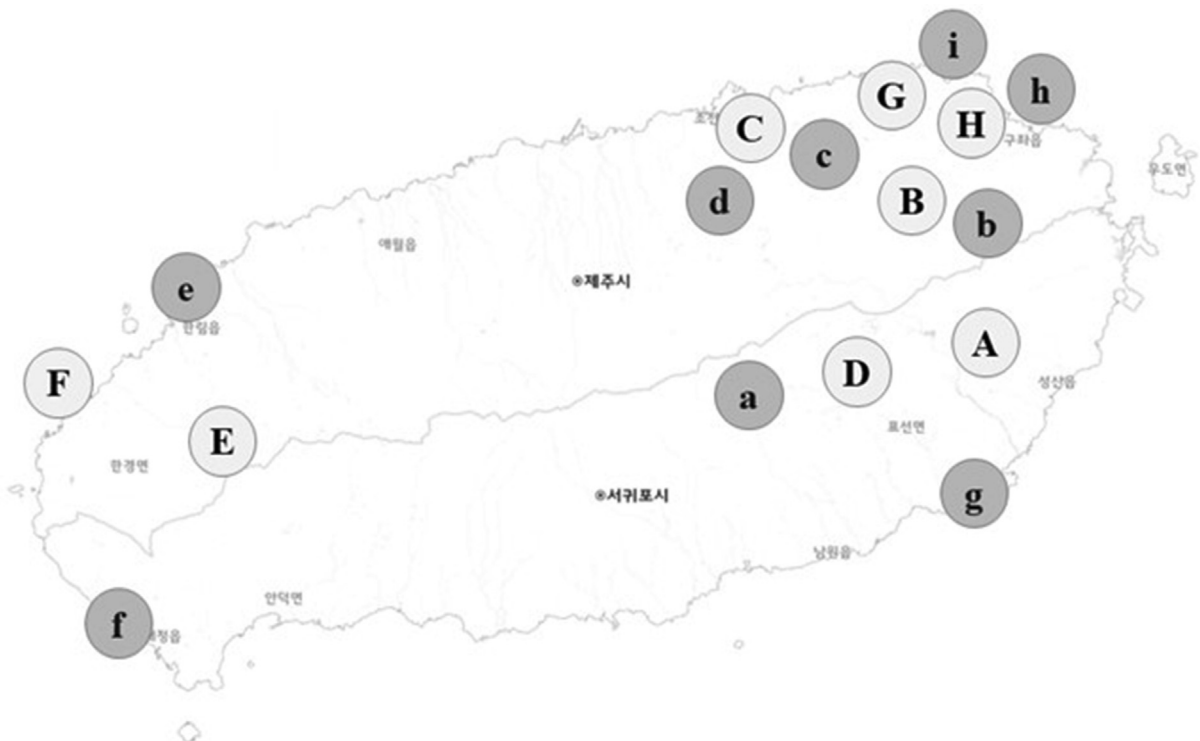


Fig. 3. Geographic locations of wind farms in Jeju Island.

minimize the error between the predicted and observed data. The universal kriging function is defined as [41].

$$Z(\mathbf{s}) = \sum_{i=0}^m \beta_i x_i(\mathbf{s}) + \varepsilon(\mathbf{s}) \quad (10)$$

where $z(\mathbf{s})$ is the capacity factor of the target point, and $\mathbf{s} = (s_1 s_2)'$ is a two-dimensional spatial coordinate.

In this paper, cross-validation of the wind power capacity factor at Jeju Island was performed. The root-mean-squared error (RMSE) was calculated as the verification index as follows [42]:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (11)$$

where $\hat{y}_i$ is the predicted data and y_i is the measured data of the target point.

3. Estimation of wind energy potential for power grid expansion planning

The capacity factors of eight wind farms on Jeju Island were calculated using 2017's 1-h unit wind power data clustered over the four seasons to analyze the probability models. Wind farm location data such as latitude and longitude were used to perform spatial modeling based on the probability distribution calculated via the Weibull distribution and Monte Carlo simulations. Fig. 3 shows the location of each wind farm on Jeju Island. In this figure, capital letters in white are existing points of wind farms and lower-case in grey are target points.

Before estimating the potential of wind energy through probabilistic modeling, the wind outputs data were preprocessed as capacity factors, and the capacity factor patterns of each wind power plant in Jeju were analyzed. South Korea is located in a temperate–climate zone in the mid–latitudes, with four distinct seasons (spring, summer, fall, and winter), and Jeju Island is a volcanic island that has stronger wind than inland South Korea. Wind speed has a high correlation with wind power output, and if the geographical characteristics of Jeju Island are considered, the wind power output is expected to be highest in winter. As a result of analysis through case study, the wind power capacity factor was highest in winter and lowest in summer. Fig. 4 shows the seasonal wind power capacity factor pattern.

Fig. 5 shows the proposed probabilistic model to estimate wind energy potential for power grid expansion planning. As shown in Fig. 5, the potential capacity factors for new wind farm locations can be determined by the proposed probabilistic model. The algorithm consists of four steps, and the contents of each step are as

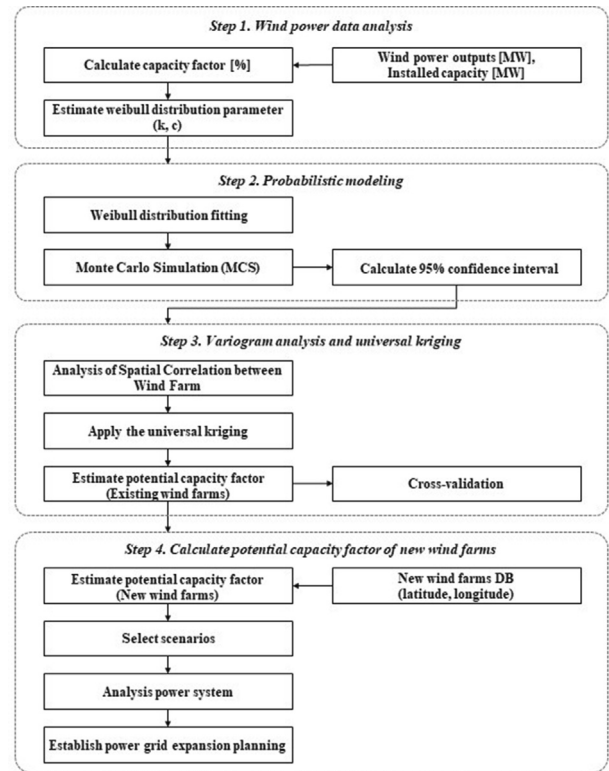


Fig. 5. Proposed probabilistic model for estimating wind energy potential.

follows. Step 1 is the process of calculating the wind power capacity factor based on wind power outputs and installed capacity, and analyzing the characteristics of wind power data. In this paper, data for each wind farm were analyzed by EPM and two-parameter weibull distribution. Step 2 is the probabilistic modeling process. Monte carlo simulations were performed based on the Weibull distribution parameters calculated for each wind farm. The maximum and minimum values of the 95% confidence interval of the extracted sampling data are used as input data to perform the universal kriging of step 3. In step 3, universal kriging was performed based on the data of eight existing wind farms, and the proposed model was verified through cross-validation. In step 4, the potential capacity factor is estimated using the location (latitude and longitude) data of Jeju new wind farms for universal kriging. The estimated data is used as a renewable energy generation input scenario of the practical power system, and power grid expansion planning can be established through the analysis of power system elements such as branch flow and bus voltage.

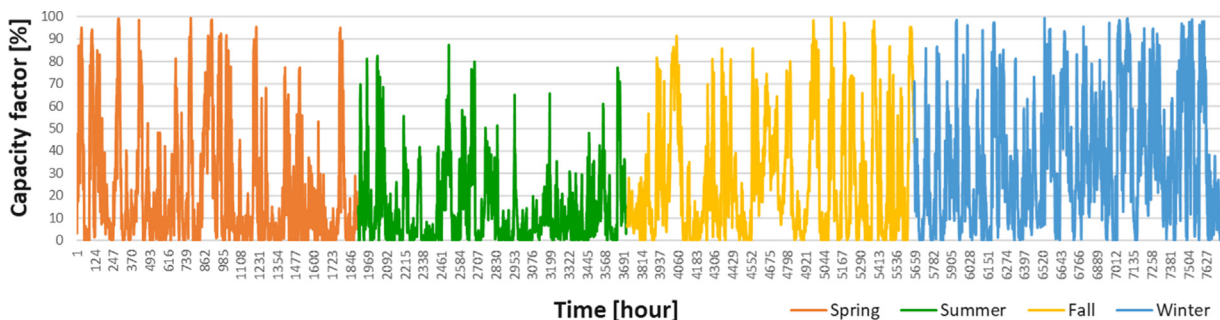


Fig. 4. Seasonal wind power capacity factor pattern for wind farm A.

3.1. Capacity factors of the wind farms based on probabilistic modeling

Table 1 reports the seasonal Weibull distribution parameters (k , c) at eight points calculated using the EPM. Monte Carlo simulations

were performed based on the two-parameter Weibull distribution. The probability densities from the Monte Carlo simulation results for wind farm A based on EPM are shown in Fig. 6.

The 95% confidence interval of each data item was calculated to compare the theoretical Weibull distribution data with the data

Table 1
Weibull distribution parameters.

Season	Parameter	Farm A	Farm B	Farm C	Farm D	Farm E	Farm F	Farm G	Farm H
Spring			1.1444	1.1028	1.1238	1.0785	1.1015	1.1391	1.1603
Summer	c [MW]	0.2323	0.3028	0.2612	0.3065	0.2114	0.1811	0.4672	0.3708
	k	1.0683	1.0826	1.0346	1.0468	1.0653	1.0815	1.0552	1.0926
Fall	c [MW]	0.1441	0.1646	0.1222	0.1636	0.1385	0.1025	0.2096	0.2214
	k	1.1803	1.1463	1.0970	1.1787	1.1685	1.2279	1.1997	1.1811
Winter	c [MW]	0.2861	0.2744	0.2546	0.3452	0.2414	0.2293	0.4442	0.3509
	k	1.3824	1.4844	1.3409	1.5656	1.4991	1.5566	1.5971	1.6952
	c [MW]	0.3910	0.4631	0.4540	0.5061	0.3961	0.3148	0.8169	0.6330

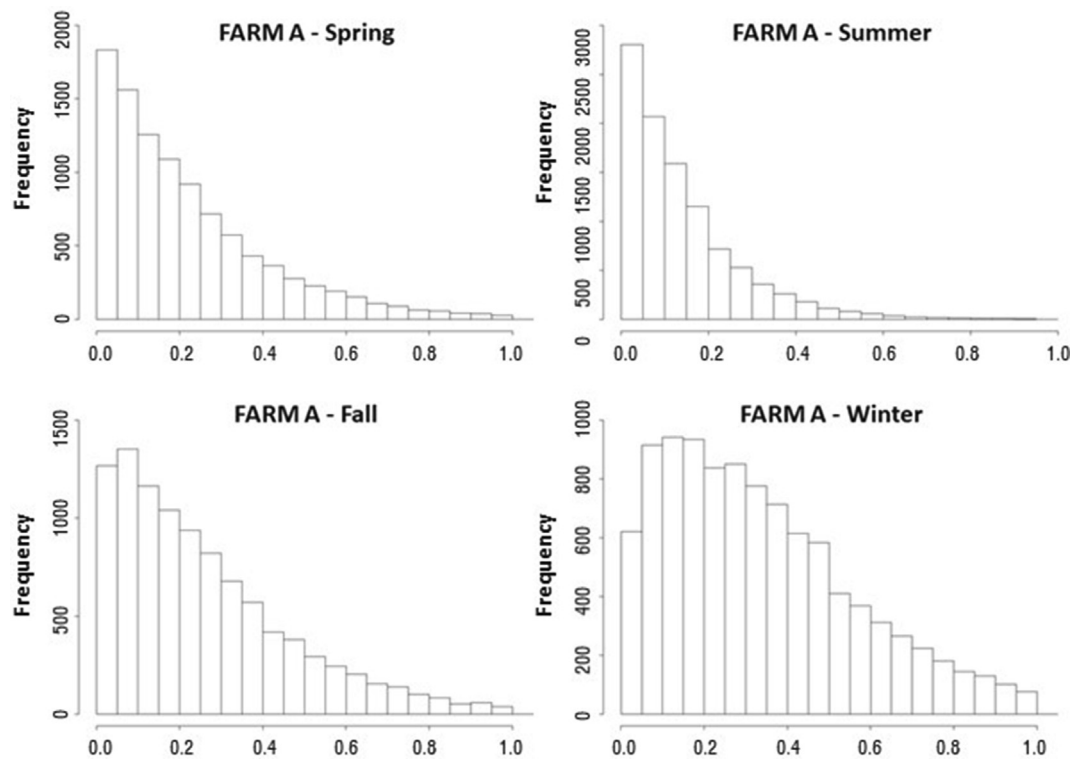


Fig. 6. Results based on Monte Carlo Simulation for wind farm A.

Table 2
95% confidence intervals and RMSE.

Season	Data	Farm A	Farm B	Farm C	Farm D	Farm E	Farm F	Farm G	Farm H
Spring	measured	[0.0076, 0.7824]	[0.0122, 0.9475]	[0.0093, 0.8533]	[0.0116, 0.9791]	[0.0070, 0.7092]	[0.0064, 0.5922]	[0.0185, 1.4696]	[0.0156, 1.1420]
	predicted	[0.0095, 0.7482]	[0.0138, 0.9250]	[0.0111, 0.8234]	[0.0135, 0.9507]	[0.0087, 0.6790]	[0.0077, 0.5713]	[0.0211, 1.4328]	[0.0174, 1.1191]
	RMSE	0.0242	0.0160	0.0212	0.0201	0.0214	0.0148	0.0261	0.0162
Summer	measured	[0.0046, 0.4891]	[0.0055, 0.5496]	[0.0035, 0.4317]	[0.0049, 0.5693]	[0.0044, 0.4717]	[0.0034, 0.3428]	[0.0064, 0.7220]	[0.0077, 0.7313]
	predicted	[0.0058, 0.4666]	[0.0068, 0.5269]	[0.0047, 0.4064]	[0.0064, 0.5387]	[0.0056, 0.4495]	[0.0042, 0.3285]	[0.0083, 0.6855]	[0.0093, 0.7034]
	RMSE	0.0159	0.0161	0.0179	0.0217	0.0157	0.0101	0.0258	0.0198
Fall	measured	[0.0127, 0.8646]	[0.0111, 0.8569]	[0.0089, 0.8369]	[0.0153, 1.0447]	[0.0104, 0.7378]	[0.0115, 0.6640]	[0.0207, 1.3187]	[0.0156, 1.0598]
	predicted	[0.0139, 0.8510]	[0.0126, 0.8369]	[0.0108, 0.8062]	[0.0167, 1.0278]	[0.0115, 0.7244]	[0.0121, 0.6591]	[0.0223, 1.3028]	[0.0170, 1.0432]
	RMSE	0.0097	0.0142	0.0217	0.0120	0.0095	0.0035	0.0113	0.0118
Winter	measured	[0.0274, 1.0052]	[0.0389, 1.1158]	[0.0293, 1.2018]	[0.0484, 1.1651]	[0.0341, 0.9461]	[0.0297, 0.7282]	[0.0818, 1.8498]	[0.0724, 1.3671]
	predicted	[0.0271, 1.0125]	[0.0381, 1.1268]	[0.0292, 1.2075]	[0.0473, 1.1766]	[0.0334, 0.9555]	[0.0290, 0.7355]	[0.0800, 1.8676]	[0.0711, 1.3784]
	RMSE	0.0052	0.0078	0.0040	0.0082	0.0067	0.0052	0.0127	0.0080

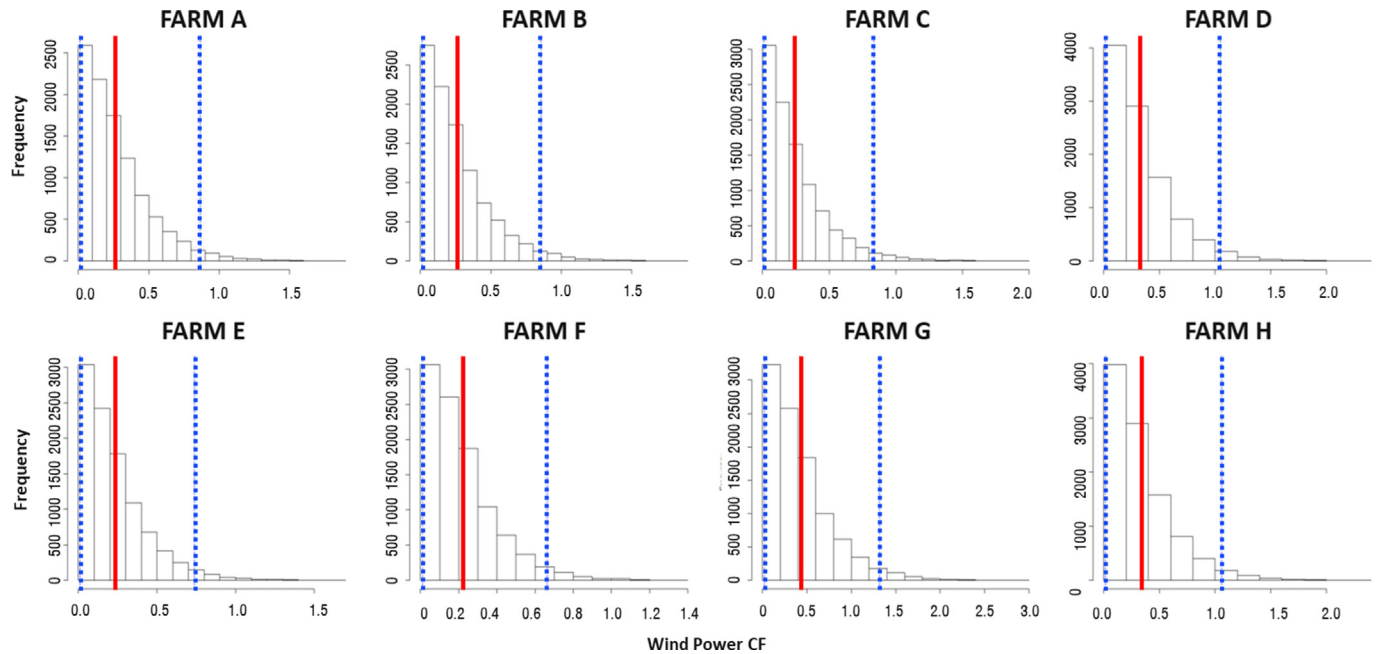


Fig. 7. 95% confidence intervals and the means of Jeju wind farms for fall.

Table 3

Cross-validation of seasonal wind farms for Jeju Island.

Season	Data	Farm A	Farm B	Farm C	Farm D	Farm E	Farm F	Farm G	Farm H
Spring	measured	[0.0095, 0.7482]	[0.0138, 0.9250]	[0.0111, 0.8234]	[0.0135, 0.9507]	[0.0087, 0.6790]	[0.0077, 0.5713]	[0.0211, 1.4328]	[0.0174, 1.1191]
	predicted	[0.0131, 0.9404]	[0.0145, 0.9884]	[0.0132, 0.8912]	[0.0034, 0.3170]	[0.0032, 0.2670]	[0.0037, 0.3908]	[0.0252, 1.5718]	[0.0114, 0.6986]
	RMSE	0.0185	0.002	0.0023	0.2009	2.89E-05	0.0163	0.0097	0.0884
Summer	measured	[0.0058, 0.4666]	[0.0068, 0.5269]	[0.0047, 0.4064]	[0.0064, 0.5387]	[0.0056, 0.4495]	[0.0042, 0.3285]	[0.0083, 0.6855]	[0.0093, 0.7034]
	predicted	[0.0062, 0.5279]	[0.0055, 0.5126]	[0.0065, 0.5412]	[0.0055, 0.3979]	[0.0006, 0.0018]	[0.0026, 0.2067]	[0.0110, 0.9143]	[0.0070, 0.5447]
	RMSE	0.0019	0.0001	0.0091	0.0099	2.55E-05	0.0074	0.0262	0.0126
Fall	measured	[0.0139, 0.8510]	[0.0126, 0.8369]	[0.0108, 0.8062]	[0.0167, 1.0278]	[0.0115, 0.7244]	[0.0121, 0.6591]	[0.0223, 1.3028]	[0.0170, 1.0432]
	predicted	[0.0152, 0.9993]	[0.0156, 1.0644]	[0.0130, 0.9130]	[0.0048, 0.4337]	[0.0111, 0.0838]	[0.0029, 0.2955]	[0.0194, 1.3573]	[0.0074, 0.6941]
	RMSE	0.011	0.0259	0.0057	0.1765	2.03E-07	0.0661	0.0015	0.061
Winter	measured	[0.0271, 1.0125]	[0.0381, 1.1268]	[0.0292, 1.2075]	[0.0473, 1.1766]	[0.0334, 0.9555]	[0.0290, 0.7355]	[0.0800, 1.8676]	[0.0711, 1.3784]
	predicted	[0.0405, 1.1917]	[0.0450, 1.2789]	[0.0355, 1.0250]	[0.0018, 0.4450]	[0.0510, 0.4750]	[0.0029, 0.5822]	[0.0876, 1.6970]	[0.0211, 0.6803]
	RMSE	0.0161	0.0116	0.0167	0.2687	0.0003	0.0121	0.0146	0.2449

extracted through Monte Carlo simulation. Table 2 summarizes the 95% confidence interval and RMSE for each season. Fig. 7 shows the Monte Carlo simulation results for the existing points in the fall. In this figure, red lines are the means and blue lines are 95% confidence intervals.

3.2. Estimation of the capacity factor of new wind farms based on spatial modeling

Cross-validation was performed by applying the 95% confidence interval of wind farms calculated from the Weibull distribution and Monte Carlo simulation results for universal kriging. Table 3 reports

Table 4

Capacity factor estimation for new wind farms for Jeju Island.

Name	LON	LAT	Spring	Summer	Fall	Winter
NEW_A	126.6681	33.36829	[0.0081, 0.6195]	[0.0047, 0.4122]	[0.0095, 0.7498]	[0.0216, 0.8253]
NEW_B	126.7684	33.51656	[0.0138, 0.9527]	[0.0058, 0.4822]	[0.0127, 0.8764]	[0.0349, 1.2187]
NEW_C	126.721	33.53811	[0.0115, 0.8436]	[0.0048, 0.4170]	[0.0110, 0.8211]	[0.0311, 1.2248]
NEW_D	126.7093	33.53121	[0.0117, 0.8443]	[0.0050, 0.4291]	[0.0109, 0.8225]	[0.0326, 1.2025]
NEW_E	126.259	33.43245	[0.0077, 0.6700]	[0.0056, 0.4662]	[0.0054, 0.5760]	[0.0249, 1.1390]
NEW_F	126.2196	33.23277	[0.0049, 0.4900]	[0.0047, 0.3980]	[0.0092, 0.6225]	[0.0190, 0.7608]
NEW_G	126.8187	33.30227	[0.0102, 0.8207]	[0.0054, 0.4725]	[0.0158, 0.9679]	[0.0287, 1.0945]
NEW_H	126.8482	33.53948	[0.0192, 1.2943]	[0.0075, 0.6284]	[0.0184, 1.1611]	[0.0625, 1.6525]
NEW_I	126.8027	33.563	[0.0170, 1.1384]	[0.0086, 0.6638]	[0.0163, 1.0491]	[0.0652, 1.4477]

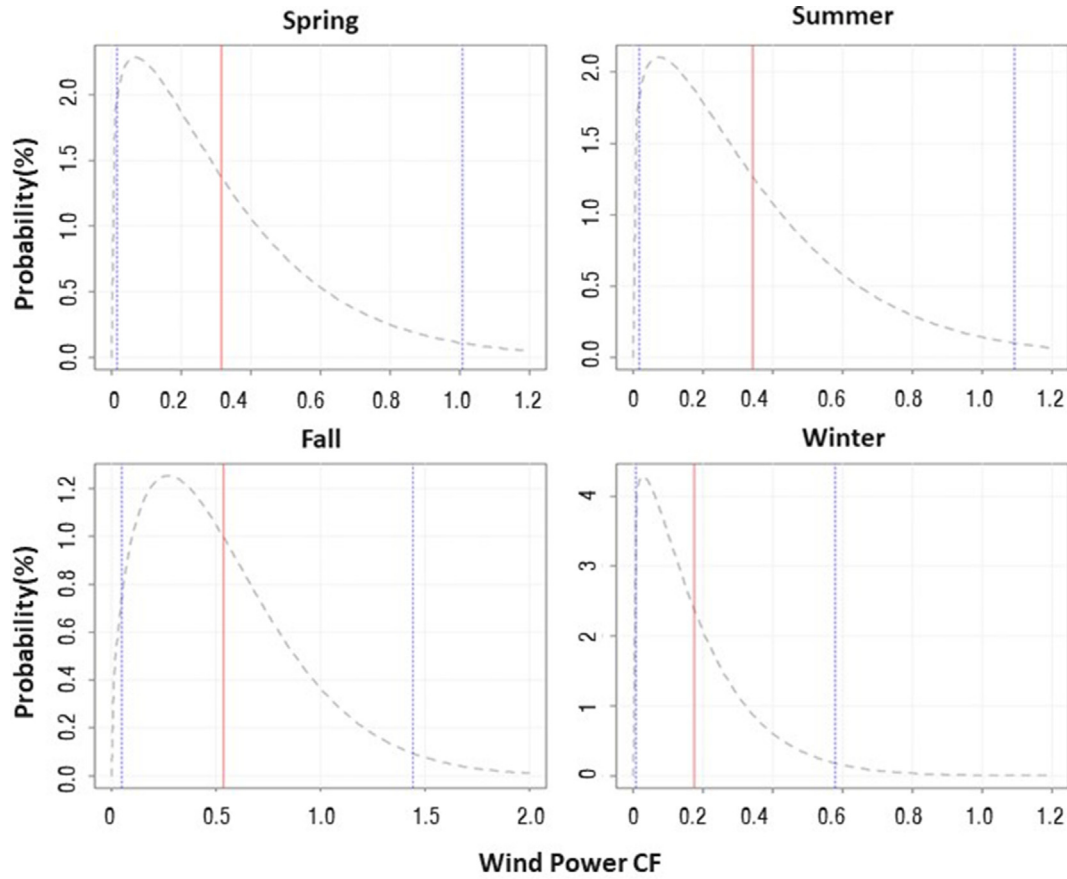


Fig. 8. Estimation of the capacity factor of new wind farm C.

Table 5

Pseudo-code of probabilistic modeling algorithm.

Algorithm Probabilistic Modeling	
Input: Wind power outputs, Installed capacity	
Output: Capacity factor of new wind farms	
Initialization $Capacity\ factor\ [\%] = \frac{Wind\ power\ output\ [MW]}{Installed\ capacity\ [MW]} \times 100;$	
while Probabilistic modeling do	
for all $v \in Capacity\ factor\ [\%]$ do	
if select EPM method then	
$E_{pf} = \frac{v^3}{(\bar{v})^3};$	
end if	
if fitting Monte Carlo Simulation (MCS) then	
$k = 1 + \frac{3.69}{(E_{pf})^2};$	
$c = \frac{\bar{v}}{\Gamma(1 + 1/k)};$	
$f(v) = \left(\frac{k}{c}\right) \left(\frac{v}{c}\right)^{k-1} \exp\left(-\left(\frac{v}{c}\right)^k\right);$	
$R(v) = e^{-\left(\frac{v}{a}\right)^\beta};$	
end if	
end for	
while Spatial modeling do	
if select universal kriging then	
$Z(\mathbf{s}) = \sum_{i=0}^m \beta_i x_i(\mathbf{s}) + \varepsilon(\mathbf{s});$	
end if	
if $x_{capacity}^{new} \in New\ wind\ farms\ DB$ then	
$Z_{new}(\mathbf{s}) = \sum_{i=0}^m \beta_i x_i(\mathbf{s}) + \varepsilon(\mathbf{s});$	
end if	
analysis power system security;	
end while	
end while	
while Power system security analysis do	
if select scenarios then	
analysis power system security;	
end if	
end while	

Table 6
Line overloading rate based on power grid scenarios.

Scenario	Line overload $80\% \leq x < 100\%$	Line overload $100\% \leq x < 150\%$	Line overload $x \geq 150\%$
Base case	Line C: 94% Line D: 86% Line H: 87% Line K: 91%	Line G: 111% Line I: 132% Line J: 148%	Line A, B: 150% Line E, F: 186% Line L: 202%
Scenario 1	Line A, B: 85% Line D, K: 83% Line E, F: 82% Line G, L: 84%	Line I: 100%	Line A, B:
Scenario 2	Line I: 87% Line J: 88% Line L: 86%	—	—

the measured and predicted cross-validation and RMSE over four seasons for the wind farms on Jeju Island with a probability of a capacity factor distribution at the 95% confidence interval.

Cross-validation assumes that one of the existing eight wind farms does not know wind farm capacity factor, and performs universal kriging based on the seven wind farms capacity factors. Cross-validation is performed on eight wind farms. To analyze the cross-validation result, the 95% confidence interval of the measured and predicted data through universal kriging were compared. The prediction error of the measured and predicted data for each wind farm was calculated through RMSE. As a result of the prediction, it was confirmed that the wind farm capacity factor prediction was performed well through universal kriging.

The reliability of the model estimation results was verified using cross-validation. The capacity factor of the new wind farm was estimated using the probabilistic model. The location of the new wind farm was set as the point where the wind resources were expected to be high [43]. Table 4 reports the potential capacity factor of the area calculated through universal kriging. A table of values means the estimated capacity factor of the new wind farm at the 95% confidence interval. Fig. 8 shows estimates of the capacity factor of the new wind farm C for different seasons. In this figure,

red lines are the means and blue lines are 95% confidence intervals.

3.3. Application of the proposed methodology

The proposed probabilistic modeling algorithm for estimating wind energy potential can be applied to all spatiotemporal scales. Table 5 shows the pseudocode of the proposed algorithm. The measured wind power outputs and installed capacity of wind farms are used to estimate the capacity factor of the new wind farm. The algorithm consists of two steps, the first step is probabilistic modeling, and the second step is power system security analysis.

In this paper, the Weibull distribution, which is the most suitable probability model for wind power output characteristics, was selected as the fitting model. The proposed algorithm can be applied to other cases with different characteristics if the data is mounted on an appropriate probability model according to the characteristics of the data to be used by the user as well as the wind power data.

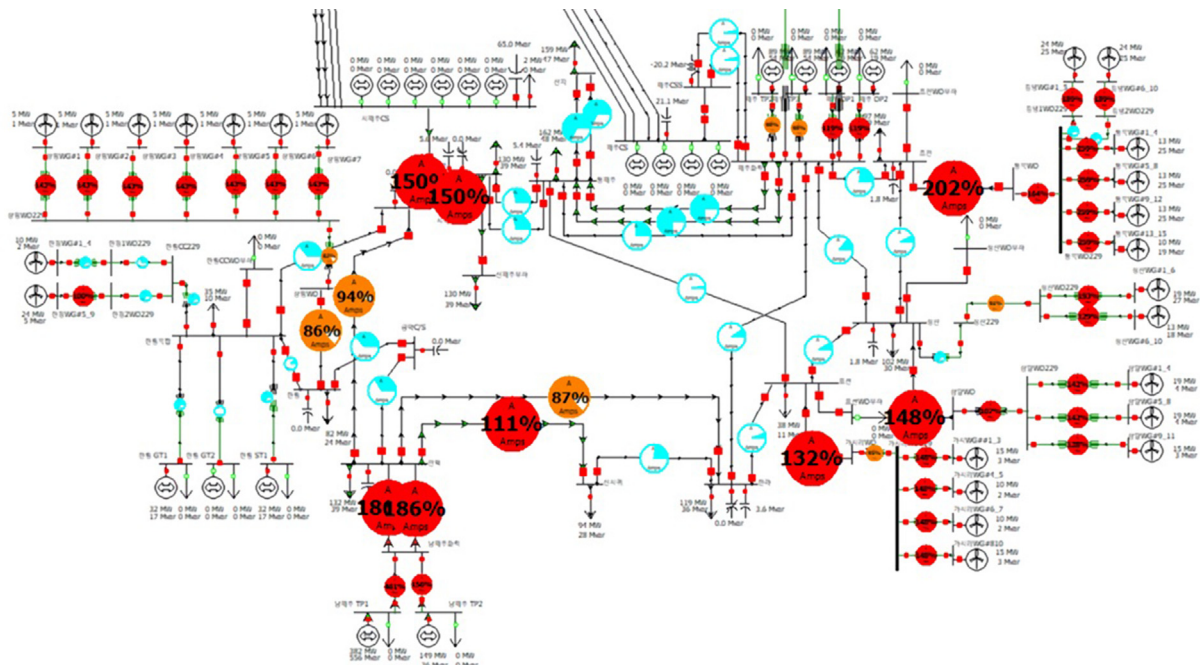


Fig. 9. Power grid expansion planning simulation for base case scenario.

3.4. Case studies: power grid expansion planning in Jeju Island's systems

The 2030 Jeju power grid was modeled using the capacity factor and 95% confidence intervals of the target points extracted through probabilistic modeling. The predicted data of peak power demand and power generation, excluding wind power, were the inputs, and the base case and two scenarios were set for the analysis of line overload and power grid expansion planning. Scenario 1 expanded the line capacity by the overload rate for lines with a load overload rate of 100% or higher than the base case, and inputs were the 95% confidence intervals of nine new wind farms estimated through probabilistic modeling. In Scenario 2, the capacity of the line with an overload rate of 80% or more of the base case was expanded by the overload rate, and the power generation of the new wind farms were the same as in Scenario 1. Table 6 reports the line overload for

each scenario.

Fig. 9 is the base case for Jeju's power grid. There is a total of eight lines where overload occurs; five have a line overload rate of 150%, and three have a line overload rate within the 100%–150% interval. In addition, there are four potential overloads (over 80% of the line overload rate). Fig. 10 shows the power simulation results of Scenarios 1 and 2 with 95% confidence intervals of the nine wind farm target points estimated in this paper. For Scenario 1, there were no lines with an overload rate exceeding 100%, nine lines with an overload rate of within the 80%–100% were analyzed. In the case of Scenario 2, there were no lines with an overload rate exceeding 100%, three lines with an overload rate of within the 80%–100% were analyzed.

As a result, Scenario 2 was more relieved of line overload on a base case than Scenario 1. To analysis power system security, predict the power generation of new wind farms and put them into the

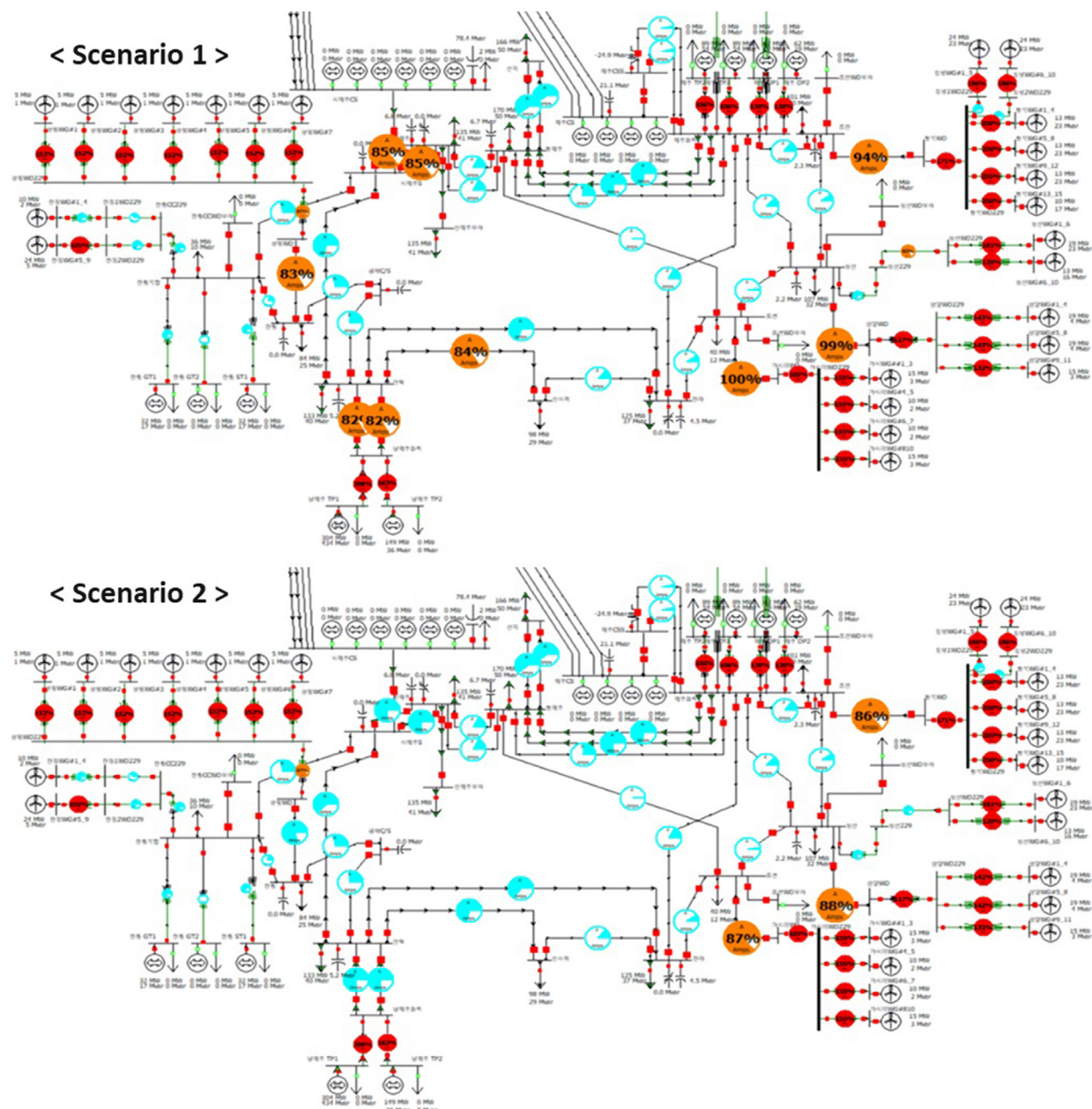


Fig. 10. Power grid expansion planning based on two scenarios.

system through the probabilistic modeling of wind energy potential proposed in this paper. In future, that probabilistic modeling will be used in future system expansion plans.

4. Conclusions

The increasing integration of wind energy generation into energy systems has led to difficulties in power flow calculations due to uncertainty and variability, which significantly affect the stability and reliability of power grids. In this paper, we propose the probabilistic modeling of wind energy potential for power grid expansion planning. The probabilistic models that considered the correlations of wind farms were implemented using wind power output data from Jeju Island, Korea, and the capacity factors of wind farms were estimated. To estimate the capacity factor of the new wind farm through universal kriging, power output data from Jeju Island's wind farms were calculated as capacity factors, and 95% confidence intervals and means were sampled by seasonal Weibull distribution and Monte Carlo simulations. To perform Monte Carlo simulations, the optimal probability model with the lowest BIC was selected, and each parameter (k , c) of the Weibull distribution was calculated through the EPM. Sampling data from probability models were used for spatial modeling, and universal kriging was performed by applying a variogram based on the spherical model and specific parameters considering the existing wind farms. Cross-validation was performed to estimate the capacity factors of the new wind farm locations and to verify the reliability of the model's estimated results. Cross-validation results were quantitatively evaluated using RMSE through universal kriging. Based on the estimated capacity factors of the target points calculated using the Kriging technique, we analyzed the potential of wind resources through system analysis and formulated the power grid expansion plan. As we discussed in previous sections, the wind energy potential estimation is needed to select the optimal location for new wind farm candidate sites. The potential capacity factors of wind generating resources at the target location can be applied to the calculation of operational reserves for enhancing the power system reliability and power grid expansion planning for high wind power penetrations.

In future, we will use the proposed algorithm to create the probabilistic renewable energy penetration scenarios, and will deploy it to enhance the security of power grids.

Credit author statement

GYEONGMIN KIM: Data curation, Writing – original draft, Software, Visualization, Investigation. JIN HUR: Conceptualization, Methodology, Software, Supervision, Writing- Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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